HYDRAULIC VARIABLE VALVE TIMING ACTUATOR [SKYACTIV-G 1.5 (WITH 4-2-1 EXHAUST SYSTEM)]

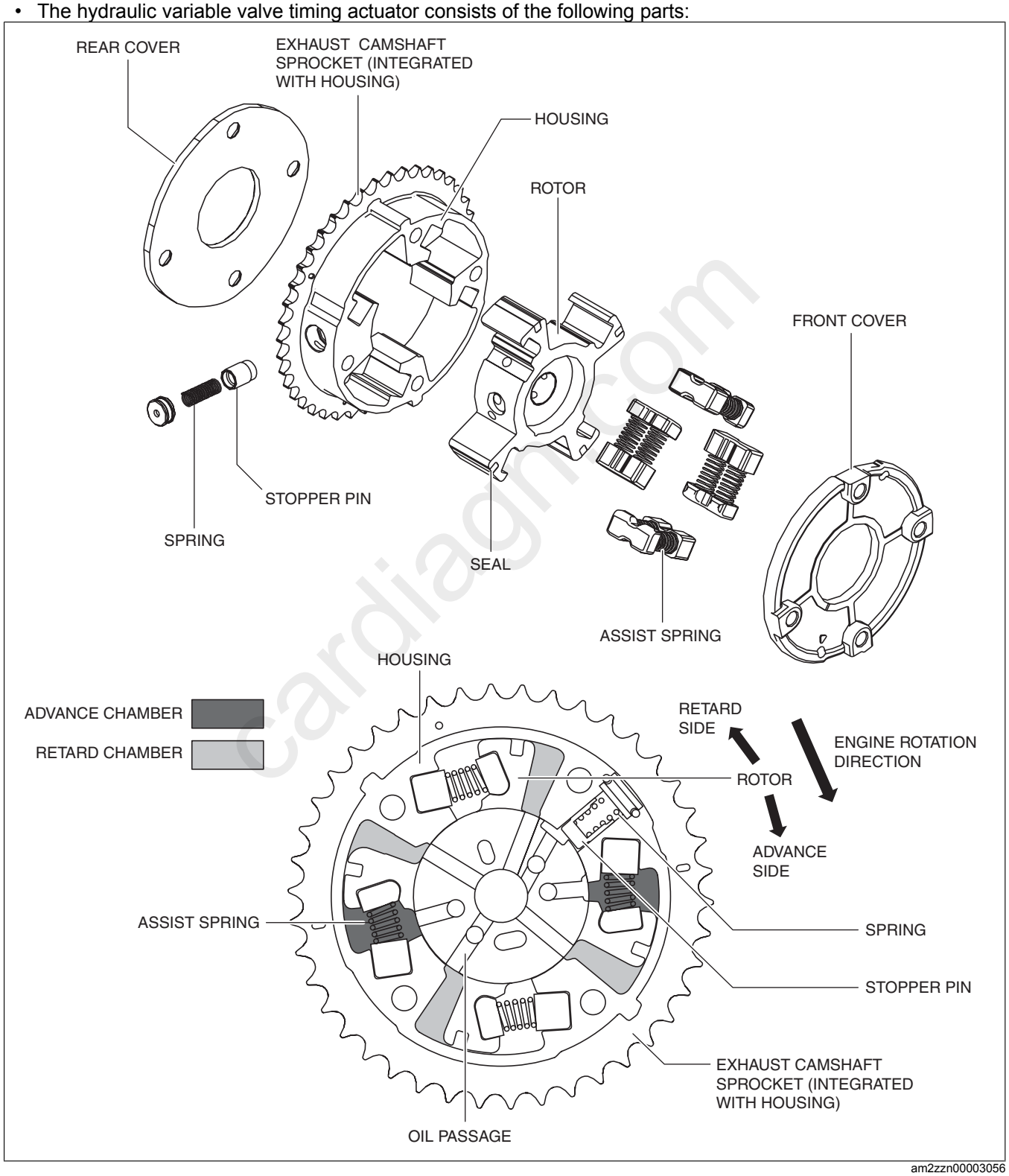
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Purpose, Function

- The hydraulic variable valve timing actuator operates using the engine oil supplied by the OCV, and changes the phases of the exhaust camshaft (valve timing).
- The hydraulic variable valve timing actuator cannot be disassembled because it is a precision unit.

Construction

- The hydraulic variable valve timing actuator consists of the following parts:



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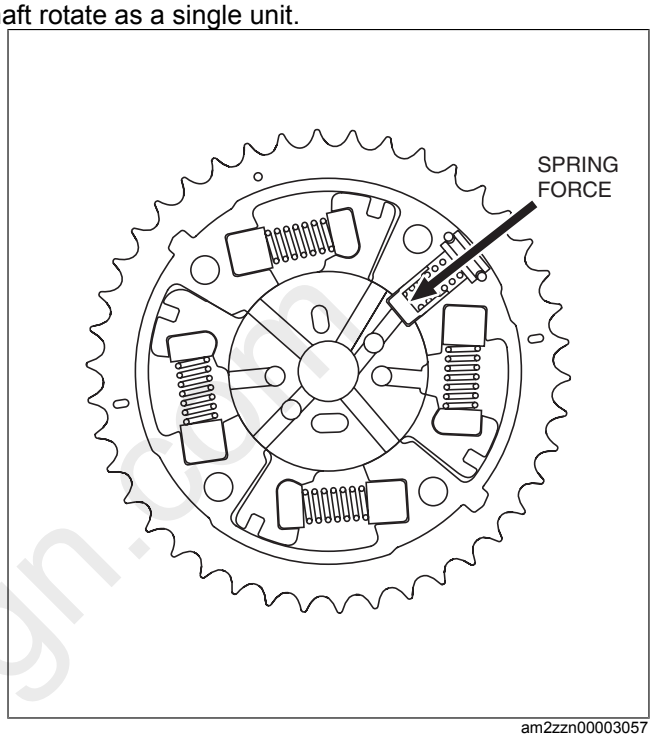
— The exhaust camshaft sprocket is integrated with the housing.
— The rotor and the exhaust camshaft are engaged and they rotate together.

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— There are two types of hydraulic chambers; advance chamber and retard chamber, between the housing and the rotor.
— When the rotor (exhaust camshaft) is in the maximum advance position, the stopper pin is engaged with the rotor depression.
— The assist spring assists to rotate the rotor in the advance direction in advance mode.

Operation
Fixed mode

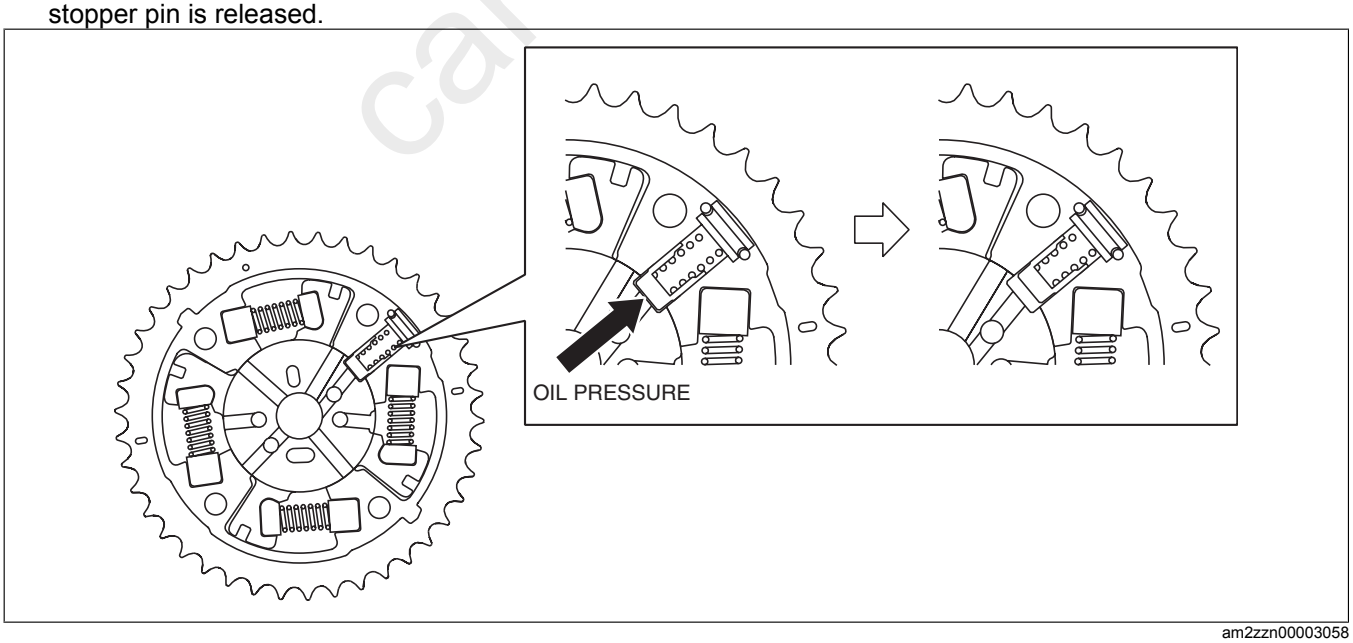
- The stopper pin inside the hydraulic variable valve timing actuator is engaged with the depression of the rotor positioned in the maximum advance position by the spring force.
- The exhaust camshaft sprocket and the exhaust camshaft rotate as a single unit.



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Retard mode

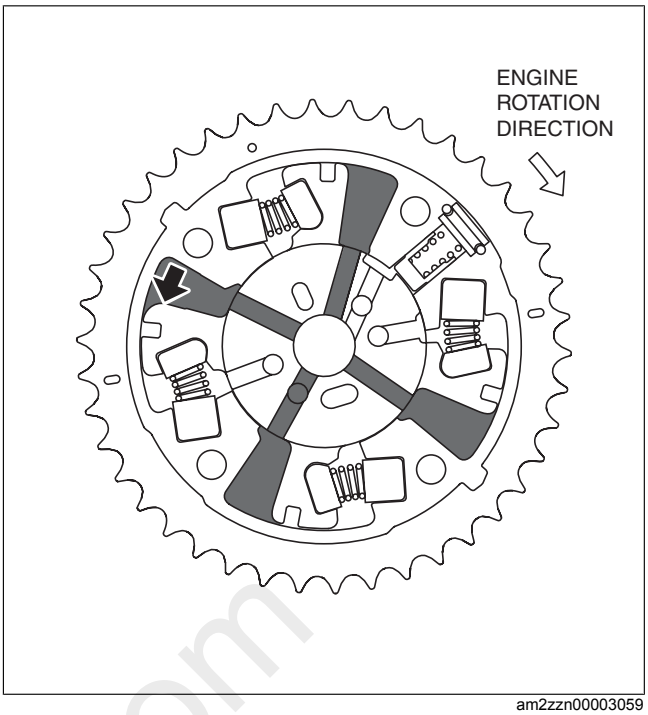
- Engine oil is led to the oil passage by the OCV, and the oil pressure is applied to the stopper pin. As a result, stopper pin is released.



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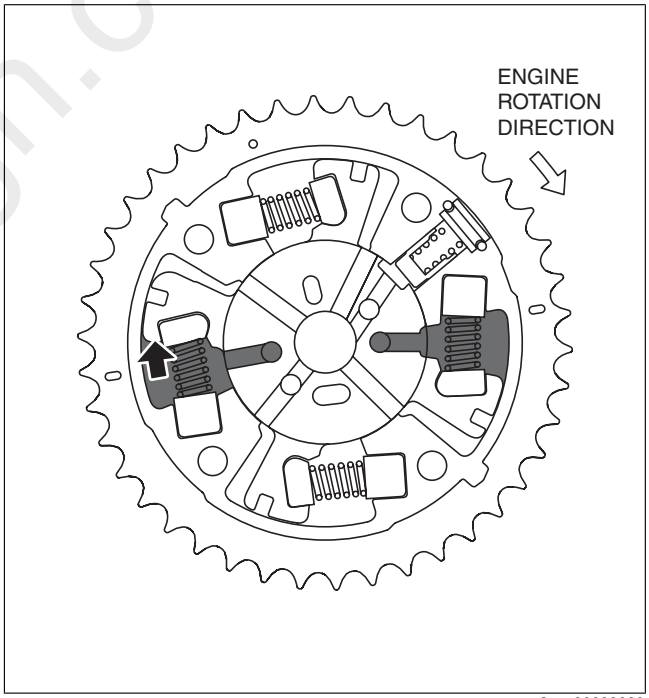
- When the stopper pin is released, oil pressure is applied to the retard chamber simultaneously. At this time, the spring force of the assist spring creates resistance, however due to the high oil pressure, the rotor rotates in the retard direction relative to the exhaust camshaft sprocket and the valve timing is retarded.



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Advance mode

- The oil passage is switched by the OCV, and along with the oil pressure being applied to the advance chamber, the spring force of the assist spring assists the rotation of the rotor. As a result of this oil pressure and spring force, the rotor rotates in the advance direction relative to the exhaust camshaft sprocket, and the valve timing is advanced.

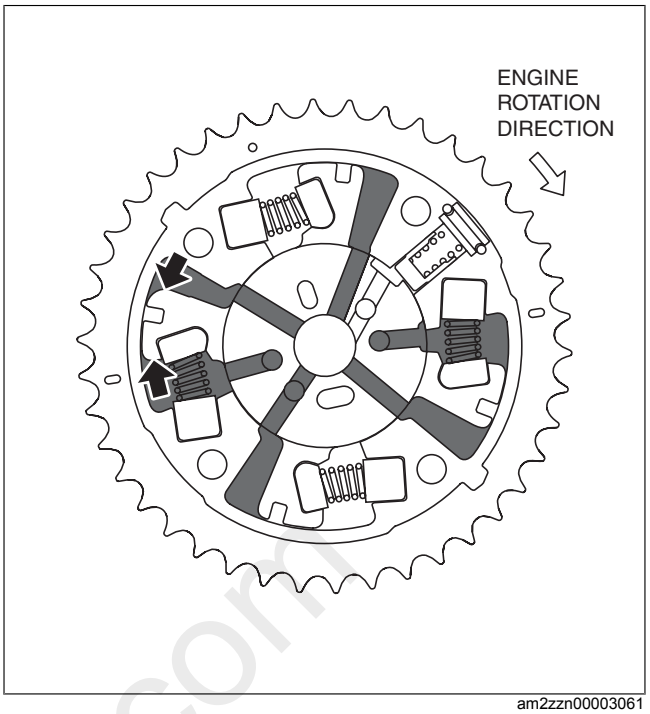


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Hold mode

- The oil pressure is applied from the OCV to both the advance and retard chambers. As a result, the relative angle of the rotor and the intake camshaft sprocket is not changed and the valve timing is held.



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